Historical Verb Morphology in Toda

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Introduction

Ethnography

- Small group of around 1,600 people living in the Nilgiri Hills of Tamil Nadu
- In close contact with Baḍaga, Kōta, and Kuṟumba speakers
- Live in small communities called *moḏ* (DEDR 4777, **maṉṯ*)
- Central component of culture is the buffalo

Documentation

- Emeneau (1941-1984)
- Bhaskararao-Tsuyoshi (2001)
- Sakthivel (1977)
- Pope (1873)

Phonology

- Seven vowel qualities: [a], [i], [ï], [u], [ü], [e], [o], [ö]
- Consonants:

	velar	retroflex	alveolo-platal and palatal	alveolar	post-dental	dental	labial
stop and affricate	p b	t d	c ʒ	t̲ d̲	č j	ṭ ḍ	k g
nasal	m			n		ṇ	
fricative and trill	f	θ		r̲		ṛ	x
lateral				ɫ l		ɫ̣ ḷ	
sibilant			s	s̲	š ž	ṣ ẓ	
trill			r				
continuant					y		w
sibilants added in fast speech			z	z̲			

Sound Changes: Assumptions

1. *a, *i, *u, *e, *o > o, ï, ü, ö, wï
2. **#_Ca/#_Ce | *a, *i, *u, *e, *o > id. except [wa]**
3. *k, *c, *ṭ, *t, *p, *t̲ > x, t_i/s_m, ṛ, θ, p/f, r̲
4. **N_# | *k, *c, *ṭ, *t, *p, *t̲ > g, ʐ, ḍ, d, b, d̲**
5. P_i_# | *k, *c, *ṭ, *t, *p, *t̲ > id. except [ɕ]; degem.
6. #_i | *cc, *ñc > c, j
7. ***r > r, š, ṣ, Ø; *ẓ > ḍ, ṛ, š, ṣ, Ø**
8. *l > l, s̲, ɬ; *ḷ > ḷ, ɬ̣
9. *y > y; *w > p_i, f_m

Sound Changes: Rule 2

- Examples
  - *amunkŭ > **o**mg- | *naṭa/*naṭe > n**a**ṛ-
  - *tirŭmpŭ > t**ï**rb- | *nīṭŭ > n**i**·ṛ-
  - *cuṯṯŭ > t**ü**ṯ- | *cuṭŭ > t**u**ṛ-
  - *keṭŭ > k**ö**ṛ- | *neṯa/*neṯe > n**e**ṛ-
  - *koy > k**wï**y- | *oravŭ ~ *orevŭ > w**a**rf-
- Exceptions
  - *aṯi (or *aṯay) > **a**ṛy
  - *tika/*tike (or *tikŭ) > t**ï**x-
  - *cuẓi (or *cuẓay) > t**u**ḍy-
  - *eṯi (or *eṯay) > **e**ṛy-
  - (*orŭnkŭ/*wānkŭ) *ōnkŭ > ***wa**·g-
- Pope: *oṛa[n?]kŭ > v**o**r**g**-m-a

	*a	*i	*u	*e	*o
#_Ca	***a***	i	u	e	***wï***
else	***o***	ï	ü	ö	***wa***

Sound Changes: Rule 4

- Examples
  - *ākŭ > o·**x**- | *ponkŭ > pï**g**-
  - *kūcŭ > ku·**s**- | *añcŭ > o**ʒ**-
  - *keṭŭ > kö**ṛ**- | *iṇṭŭ > ï**ḍ**-
  - *poṯ > pï**ṯ**- | *tōnṯŭ$_{PST}$ > twï·**ḏ**-
  - *catŭkŭ > to**θ**x- | *kintŭ > kï**d**-
  - *opŭcc > *uvŭcc- > u**f**c- | *tumpŭ > tü**b**-
- Indications
  - *oṇŭkŭ > wïṇ**x**- | *oṇŭnkŭ > wïṇ**g**-
  - *u[y?]ar-t- > u·**θ**- | *u[y?]ar-nt- > u·**d**- (u·r- for both)
  - *wār-t- > po·**θ**- | *wa[r]-nt- > po**d**- (po·r- for both)

Emeneau: *NP ⟶ B [⟶ F]

	*k	*c	*ṭ	*ṯ	*t	*p
N_#	g	ʒ/j	ḍ	ḏ	d	b
V_V	x	s	ṛ	ṯ	θ	*w > f

What's new in today's talk?

- Goal of definitively creating reconstructions for Pre-Toda for it to be better implemented in comparative studies
- Begin with verbal bases and finite verbs, and rope in other forms later
- Project ongoing…

Verbal Bases

S_1 and S_2

- Emeneau (1967) initially posited the S_1-S_2 model for South Dravidian, where S_1 usually represents the **nonpast**, and S_2 represents the **past**
- Toda works differently from most of Dravidian, and the use of S_1 and S_2 does not always correlate with tense
- Each verb has both stems:
  - S_1 is derived from bare bases (with the occasional causative)
  - S_2 usually features a suffix that can be traced back to one of Krishnamurti's PDr past suffixes added to the base

Phonological conditioning in S_2

- Palatalization
  - S_1ċ + -y- ⟶ S_2c | *āṟŭccŭ (*āṟŭcc-i-) > o·ṟc- (o·ṟc-)
  - S_1ʒ + -y- ⟶ S_2j | *añcŭ (*añc-i-) > oʒ- (oj-)
  - S_1y + -θ- ⟶ S_2s | *oṭi (*oṭi-T- > *oṭi-C-) > wïṛy- (wïṛs-)
  - S_1y + -t- ⟶ S_2c | *oṭi (*oṭi-tt- > *oṭi-cc-) > wïṛy- (wïṛc-)
- Assimilation
  - S_1r + -θ- ⟶ S_2θ | *ir (*ir-T-) > ïr- (ïθ-)
  - S_1y + -t- ⟶ S_2t | *cay (*catt-) > soy- (sot-); cf. Ka. sāy (satt-)
  - S_1l + -d- ⟶ S_2d | *sōl (*sōt-)$_{L(Baḍ)}$ > swï·l- (swï·t-)
  - S_1l + -t- ⟶ S_2t | *sal (*sand-)$_{L(Baḍ)}$ > sal- (sad-)
  - S_1y + -d- ⟶ S_2d | *wey (*weT-) > pöy- (pöd-)
- Sandhi
  - S_1ṟ/ṛ/ḍ/x + -θ-/-t-* ⟶ S_2ṭ/ṭ/ṭ/k | *pukŭ (*puk-k-) > pux- (puk-)
  - S_1ɬ/ɬ̣ + -t-* ⟶ S_2ṭ/ṭ | *kēḷ (*kēṭ-ṭ-) > ke·ɬ̣- (ke·ṭ-)
  - S_1r/l/n/ṣ/ḷ/ṇ + -θ-/-d-* ⟶ S_2d/ḍ/ḍ/ḍ/ḍ/ḍ | *uṇ (*uṇ-ṭ-) > uṇ- (uḍ-)

Extracting tense suffixes

- Four tense suffixes: -t-, -θ-, -d-, -y-
- -y- and Tamil-Kannada -i-
- -t- and Tamil-Irula -tt-
- -θ- vs -d-?
  - Emeneau suggests *-nt- > -d- > -θ-
  - Patterning with Tamil
    - Most liquid-final verbs take -nt-, 10 clear -t- pasts
    - aḻu, uḻu, toḻu, poru; ey, koy, cey, ney, pey, vai (all +t)
  - Patterning with Kannada
    - Most liquid-final (+more) verbs take -d-, 5-6 clear -nd- pasts
    - bār (band-), tār (tand-), bē (bend-), mī (mind-), nō (nond-), āṭar (āṭand-)

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Extracting tense suffixes

- Four tense suffixes: -t-, -θ-, -d-, -y-
- Pre-Toda: *-tt- (< *-T-t- or *-t-T-), *-t-, *-nt-, *-i-

- *-nt- vs. *-t- past (difference in meaning?):
  - *uyar, *uyar-t- > u·r-$_1$ (u·θ-) “disappear, belost”
  - *uyar, *uyar-nt- > u·r-$_2$ (u·d-) “disappear suddenly by magic”

S_2: notable examples

- CV̆Cŭ can take -y- past
  - *aẓ (*aẓ-i-) > oṛ- (oṛy)
  - Cf. Koḍ. or (ori-),
    OTa. koḷ (koṇṭ-, koḷi-) koḷī⁺ya {Neṭu: 144}
- CV̄Cŭ can take -t- past
  - *āṯŭ (*āṯ-ṯ-) > *āl (*āṯṯ-) > a·ɬ- (a·ṯ-)
  - Cf. Ka. bisāḍu (bisāṭ-), nāṯu (nāt-)

Verbal augments in Pre-Toda

- Likely not productive, but some nuanced meaning
  - *oṇŭkŭ > wïṇx- “wither, dry (intr.)”
  - *oṇŭnkŭ > wïṇg- “be scorched by sun”
  - *oṇŭkkŭ > wïṇk- “wither, dry (tr.)”
- Causatives
  - *-p- [> *-w-] | *peyar-vŭ > pö·šf-
  - *-cc- | *tappŭ-ccŭ > topc-
  - *-tt- | *poraṭ-ṭŭ > pï·ṭ- (cf. pï·ḷ-)
  - pu·f- “wear cloak,” pu·ṭ- “tie around someone else’s neck” and pu·c- “put garment on someone else”
  - Cf. Iruḷa indirect vs. direct

Possible non-finite forms?

- Past converb
  - ojuṟt- “terrify”
  - *añc-i > oj- “having feared”
  - *uṯŭttŭ > -uṟt- “ct experience”
  - Wilden (2020): “30% of early Tamil absolutives *change* the subject”
- Nonpast converb?
  - aṟynbeṯ- “be mad” | Emeneau: “be deranged in understanding”
  - *aṯay-n-a ~ *aṯi-n-a > aṟyn- “while knowing”
    - Possibility 2: late loan from Ta. aṟintu
    - Emeneau: noun? Cf. mu·n “face” < *mokan
  - *weṯṯ- > *peṯ- “be deranged”
  - OTa. aṭukkuṇa varinum “if they appear in sequence [i.e. while stacking]” {Tol: 718}

Finite Verbs

The Present-Future

- S_2-p-PNG (in 1, 2sg)
- *wār: *wantŭ-pp-PNG
- Why -a- > -ŭ-?
  - Possibly first-syllable stress
  - Cf. Ta. māttiram "only" > To. *māttirŭm > mo·tïrm (not *mo·tirm)
- Cf. OKa. bandappen "I come" > Ka. band^{h}ēnu, idd^{h}ēne
  - Baḍ. bandane "id." < *band-hane < *band-apen
  - Sōl. banne "id." < *bandane << *band-apen

	po·r- "come"	tïrx- "turn [intr.]"
1sg	po-d-p-en	tïrx-y-p-en
1pl.excl	po-d-p-em	tïrx-y-p-en
1pl.incl	po-d-p-um	tïrx-y-p-um
2sg	po-d-p-y	tïrx-y-p-y

Past and Dubitative

- Past: S_2-šp-PNG (in 1, 2sg)
  - Pope: edd-er-sh-pini "I was" = edd[e]špin-i
  - *wār: *want-ir[ŭ]ppen
  - Cf. Aranāḍan bī^{1}nd^{2}u^{3}pp^{4}e^5 "I fell" and Toda pïḍ^{1}θ^{2}š^{3}p^{4}en^5 "id."
- Dubitative: S_1-š(p)-PNG (in 1, 2sg)
  - Expected: *po·(r)špin-i "I may come"
  - *wār: *wara ir[ŭ]ppen
- ir or *er?
  - Pope suggests verb *er- in ešpin-i, but also uses ir- in iren-i, both "to be."

	Past	Dubitative$_{reg}$	Dubitative$_{irr}$
1sg	po-d-šp-en	pa-šp-in-i	tïrp-šp-in-i
1pl.excl	po-d-šp-em	pa-šp-im-i	tïrp-šp-im-i
1pl.incl	po-d-šp-um	pa-šp-um-i	tïrp-šp-um-i
2sg	po-d-šp-y	pa-šp-[y]-i	tïrp-šp-[y]-i

Tenseless and Negative

- Tenseless: S_2-PNG or S_1-n-PNG
  - Verbs that take -y-: tïrp-n-en
  - All other verbs: pod-en
  - Cf. OTa. participial -n-, Koraga nonpast -n-, and much more
  - OTe. and OTa. "tense" distinctions were aspect-based:
    - icc**ina**vānṟu ceḏuvānṟu "he who pays [a bribe] ought to be damned" {Stone at Peddamudiyam, Cuddapah. 9th CE. Lines 5-6}
    - ār-iruṇ-[n]aṭu nāḷ varu**ti** "you used to come at midnight in difficult darkness"{Kali. 141.3-7}
- Negative
  - S_1-PNG: po·ren-i
  - Cf. Ta. vārēṉ and OKa. bāren "id."

Secondary Paradigms

- pi·špen and pi·šn; pi·pen and pi·n
- Loss of -pV- may be a result of fast speech
  - Emeneau: “Paradigm II occurs when the verb is:
    1. Followed by an interrogative particle
    2. Followed by the alternative question particle -ïs
    3. Followed by a negative verb form
    4. Followed by the quotative verb ïn
    5. Preceded by an interrogative word somewhere in the sentence
    6. The first verb in an alternative question, or
    7. In imaginative games in children’s speech.”
  - UCLA Recordings
    - Prompter: (Ta.) nāṉ viṭṭ-iṭṭēṉ, “I left”
    - Speaker: pïṭšpin-i. Realized as [fɯt̪ʃpʲni] ~ [fɯt̪ʃmʲhi]

Voluntative

- S_1-k-PNG
- Pope: er-sh-k-en [= ešken] as present-future
- Compare
  - Ta. optative: varu-ka “let X come”
  - OTu. optative: kēḷŭ-k-oṁ “let us hear” {BhA 1.2.3}
  - Pa. imperative: cūṛe-k-en “let me hear”

Imperative

- 2SG: S_1
- 2PL: S_1-š (warrants *-ir)
  - OTa. col-īr “speak!” (Post-Caṅkam) {Tē. 2.2.8.3}
  - Pa. cūṛ-ur “see ye!”
- 3: S_1-mo· (warrants *-[ŭ]mā)
  - OTa. moḻi-mō “speak!” {Kali. 2.2}
  - OTa. teḷi-mē “let it be clear!” {Kali. 273.8}
- Negative: S_1-oṭ-(š)
  - *aṭṭŭ “to prevent < to shut”
  - Cf. Ta. aṭṭi “hindrance, obstacle”

Other negative forms

- Voluntative negative: S_1-š-PNG
- One possible formation:
  - Originally negative converb + voluntative, eg.
    wārā irŭ-kk-en
    āṭā irŭ-kk-en
  - Apply phonological changes
    *pa·r[a·]škin > paš-škin
    *ōṛ[a·]škin > ōṛ-škin
  - -kV- loss, as in Paradigm II
    pašn and ōṛšn
    *Note -k PNG (not expected -θ) in third person

Personal terminations

- 1SG: -n, -en, -in. < *-en, maybe *-ēn.
- 1PL.EXCL: -m, -em, -im. < *-em, maybe *-ēm.
- 1PL.INCL: -m, -um, -u. < *-om.
  - -u: *-um or *-om. cf. OTa. -t-um/-k-um, Kūī. māro mann-o
- 2SG: -y, -ty, -i
  - -y: *-ay
  - -ty: *-tti. Cf. OTa. uy-tti-[y]-ō {Kali. 63.4}
- 2PL: -š, -tš
  - -š: *-ir
  - -tš: *-ttir. Cf. OTa. cē-ṟir-ō {Kali. 268.1}
- 3: -u, -t, -θ
  - -u: ?
  - -t, -θ: *-t- ~ *-tt-

Reconstructing Pre-Toda

- What we know: (a bit of) Proto-Tamiḻ-Kannaḍa, and Modern Toda
- Tentative hypothesis on evolutionary stages:
  1. Imperfective (*-n-) vs. perfective (*-t-, *-i-, *-nt-, *-tt-) contrast. Voluntative (*-k-), dubitative (*-p-), and imperative (*-∅, *-m-V̄) moods. Single negative (*-∅-). Durative (*-t/i/nt/tt-ap-).
  2. Periphrasis forms new nonpast and past dubitatives (*ir)
  3. Imperfective, perfective ⟶ (suppletion) ⟶ tenseless
  4. Past dubitative shifts to past, and durative shifts to nonpast
  5. Paradigm II and mood-specific negatives form

Rethinking the Classification of Toda

- Shares many affinities with Kannaḍa
  - Present-future
  - Past suffix distribution
  - Initial *w- > [*b-] > p-
  - Denasalization of homorganic clusters: twï·ṛ-, Ka. tōḍu, Ta. tōṇṭu
- …but also with Tamiḻ
  - -n- in tenseless, and OTa. 2nd-person “subjunctives”
  - Palatalization in past stems
- …and with others
  - Separate voluntative/dubitative paradigm (see OTu, Parjī, Kūī, OTe)
  - Complex irrealis system (substratum?)

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